

GROAPA FINITA DE POTENTIAL. METODA TIRULUI

ORIGIN := 1

$$h := 6.625 \cdot 10^{-34}$$

$$m := 9.1 \cdot 10^{-31}$$

$$ee := 1.6 \cdot 10^{-19}$$

$$\hbar := \frac{h}{2 \cdot \pi}$$

Definim pasul (distanța dintre două noduri) egal cu 1 Angstrom

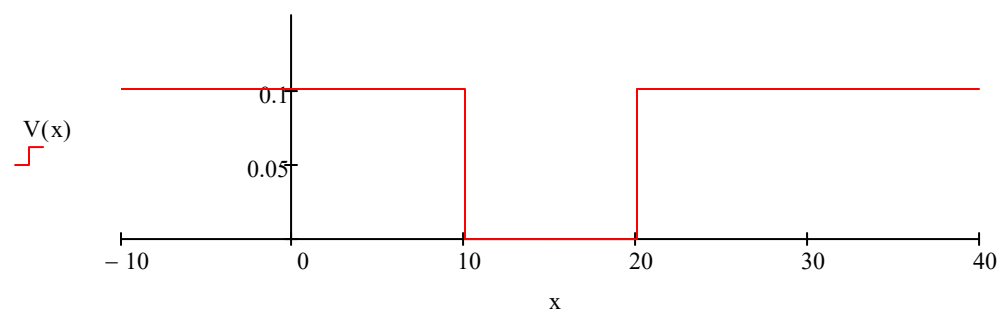
$$p := 0.1 \quad [\text{nm}]$$

Notăm cu t constanta care înmulțește paranteza (V-E)

$$t := \frac{2 \cdot p^2 \cdot 10^{-18} \cdot ee \cdot m}{\hbar^2} \quad t = 0.261926 \quad [1/\text{eV}]$$

Definim potențialul:

$$V(x) := \begin{cases} 0.1 & \text{if } x < 100 \cdot p \\ 0 & \text{if } 100 \cdot p \leq x \leq 200 \cdot p \\ 0.1 & \text{if } 200 \cdot p < x \end{cases}$$



Aplicam metoda tirului

Stabilim numarul de pasi pe axa energiei

$i_{\max} := 500$

$i := 1 \dots i_{\max}$

Stabilim pasul pe axa energiei, astfel incat energia electronului sa raman mai mica decat V:

$\text{pasE} := 0.0002$

$E_i := i \cdot \text{pasE}$

$\max(E) = 0.1$

Stabilim numarul de pasi pe axa x:

$k_{\max} := 500$

$k := 1 \dots k_{\max}$

$x_k := k \cdot p$

$\max(x) = 50$

Conditile initiale:

$\psi_{i,1} := 0$

$\psi_{i,2} := 1$

Obs: primul index indica valorile energiei iar al doilea index indica poziti

!!! solutia va fi o matrice $i_{\max} \times k_{\max}$, din care vom selecta solutiile fizice la acele valori ale energiei la care functia de unda isi schimba semnul

$n := 3 \dots kmax$

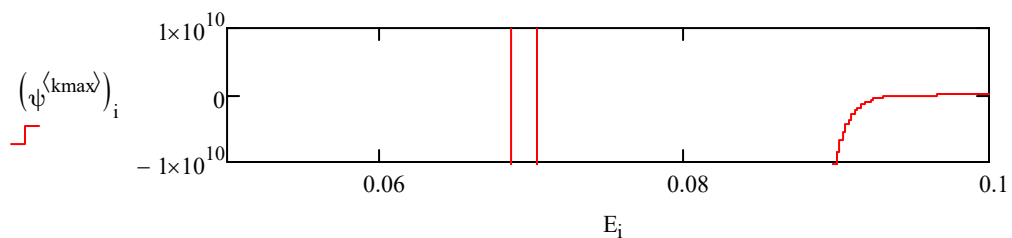
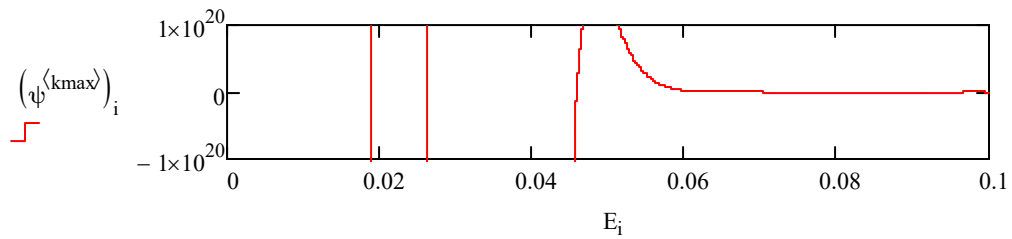
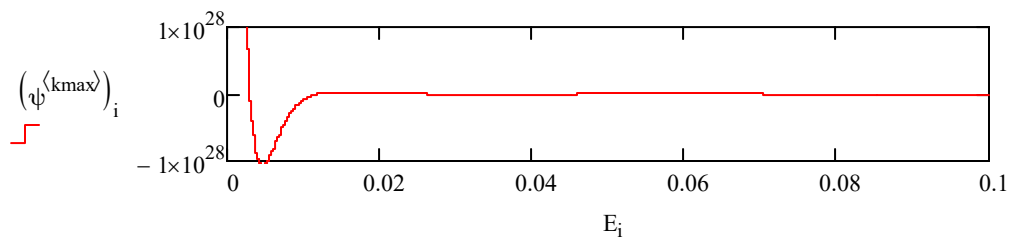
====> valorile indexului coordonata incep de la pasul 3,
pentru pasii 1 si 2 avem conditiile initiale de mai sus

$$\psi_{i,n} := \left[t \cdot \left(V(x_n) - E_i \right) + 2 \right] \cdot \psi_{i,n-1} - \psi_{i,n-2} \quad \text{====> Ec. Schrodinger}$$

Selectarea solutiilor fizice

---- analizam valorile lui ψ la distanta mare de groapa
($x \rightarrow \infty$, adica $i = imax / x = xmax$)

---- identificam valorile energiei pentru care $\psi(xmax)$ isi schimba semnul, adica intre
cele doua valori ale energiei $\psi(xmax)$ trece prin zero.



====> fiecare energie la care apare o linie verticala reprezinta o solutie fizica

Selectarea solutiilor

out := augment($E, \psi^{\langle k_{\max} \rangle}$)

	1	2
1	$2 \cdot 10^{-4}$	$2.072629 \cdot 10^{29}$
2	$4 \cdot 10^{-4}$	$1.726111 \cdot 10^{29}$
3	$6 \cdot 10^{-4}$	$1.427844 \cdot 10^{29}$
4	$8 \cdot 10^{-4}$	$1.171821 \cdot 10^{29}$
5	0.001	$9.527238 \cdot 10^{28}$
6	0.0012	$7.658529 \cdot 10^{28}$
7	0.0014	$6.070568 \cdot 10^{28}$
out = 8	0.0016	$4.726734 \cdot 10^{28}$
9	0.0018	$3.594755 \cdot 10^{28}$
10	0.002	$2.646231 \cdot 10^{28}$
11	0.0022	$1.856197 \cdot 10^{28}$
12	0.0024	$1.202738 \cdot 10^{28}$
13	0.0026	$6.666479 \cdot 10^{27}$
14	0.0028	$2.311144 \cdot 10^{27}$
15	0.003	$-1.18552 \cdot 10^{27}$
16	0.0032	...

In tabel se pot citi valorile energiei unde ψ schimba semnul (implicit avem o solutie fizica).

En :=

s ← 0	
for i ∈ 2 .. imax	
if $\frac{(\text{out}^{\langle 2 \rangle})_{i-1}}{(\text{out}^{\langle 2 \rangle})_i} < 0$	
s ← s + 1	
aux _s ← $\frac{(\text{out}^{\langle 1 \rangle})_{i-1} + (\text{out}^{\langle 1 \rangle})_i}{2}$	
continue otherwise	
aux	

En =

0.0029
0.0117
0.0261
0.0459
0.0703
0.0965